

COLOSSEUM · SOLANA

Vitals

A patient is dying on a clock and you decide what happens next.

Anyone can play. Nobody can fake the replay.

EP1 · THE LAST BITE

THE PROBLEM

Skill has no passport.

Since 2024, ECFMG certification — the gate to US residency — requires graduating from a school accredited by a WFME-recognised agency. Accreditation is institutional, leaving individual competence stranded at borders.

4,350+

Medical schools worldwide in the World Directory

~23

Countries with a WFME-recognised agency

130–147

Countries whose graduates apply annually

Local Hospital / Scored Encounters

× **Border Trust Barrier**

US Residency / Global Verification

Graduates run hundreds of scored encounters, but evidence sits trapped in siloed databases. **The evidence exists.**

The trust layer does not.

GAMEPLAY & VERIFIABLE OUTPUT

The replay is the trophy, not the score.



GHOST · MOCKUP

you · adrenaline 1:12 ghost · 0:38

Ghost racing. The tape replays exactly, so another player's run can run beside yours — and you watch the two patients diverge while you are still deciding. Trackmania's ghost car, in an emergency room.



RECORD · MOCKUP

harm · stand/walk collapse 50 saves · 12 harms

Harm is on the record. Fifty saves with twelve harms reads differently from thirty with none — and it keeps the fastest run and the cleanest run from being the same run.



DEATH CARD · MOCKUP

DeathArrest · turned at 2:14

gold path → adrenaline IM at 0:40

Failure is content. "She died because you reached for antihistamines" is a better clip than any win. A loss should be legible, teach in the same breath, and be worth showing someone.

treated adrenaline IM · O ₂ · supine · fluids · admit	WinDischarge · leaf: 65c56e...
stood up adrenaline IM · stood the patient up · O ₂ · fluids · admit	WinDischarge · harm · leaf: a6bec3...
no adrenaline antihistamine · steroid · wait	DeathArrest · fatal

Deterministic execution: Three players, three tapes, replayed against the shipped physiology automaton. Same outcome as the first, different leaf — because harm is on the record.

ONE SEASON · EP1 IS FILMED, THE REST ARE AUTHORED AUTOMATA

EP1

The Last Bite

anaphylaxis · she has minutes

EP2

Time Is Muscle

STEMI · the clock is muscle

EP3

Don't Make Him Cry

epiglottitis · doing nothing is the treatment

EP4

The Masquerader

PE · everything looks normal

EP5

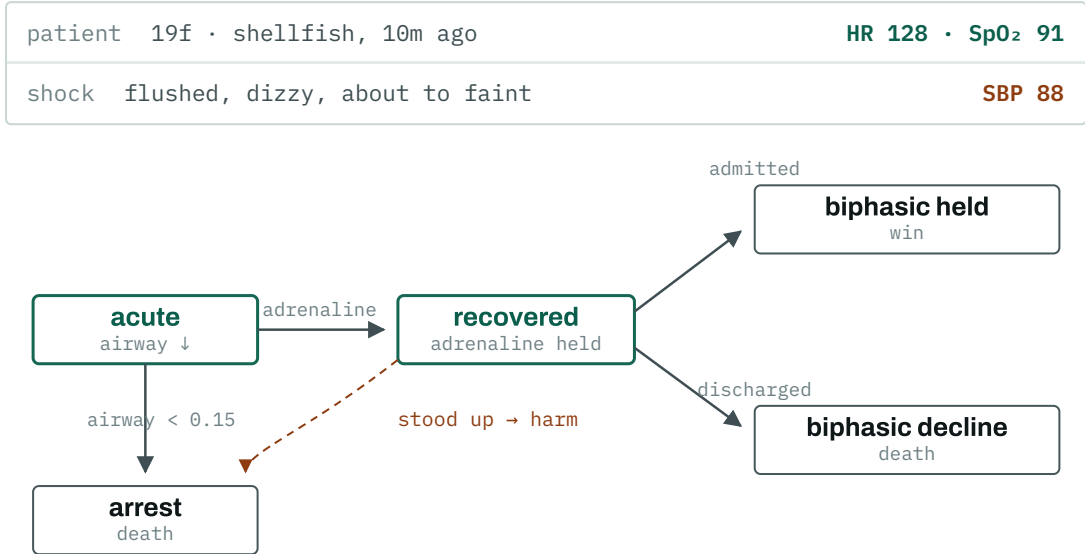
The Night the Stars Fell

mass casualty · the queue is the cost

THE MECHANISM & AUTOMATON

The model makes it playable. The automaton makes it provable.

THE SITUATION & AUTOMATON



DETERMINISTIC REDUCTION

- 1 · **commit** hash(scenario || player || nonce) onchain
- 2 · **play** Automaton tracks vitals, triggers, harm
- 3 · **reduce** Discrete facts: ordered beats, harm, outcome
- 4 · **anchor** One compressed leaf: recomputed forever
- leaf = sha256(sce_hash || tape || beats || harm || outcome)

THE AGENT

A **local model** plays her dialogue. The tape records **the question you asked, never her answer** — so history-taking is on the record and the model is nowhere near the hash.

THE AUTOMATON

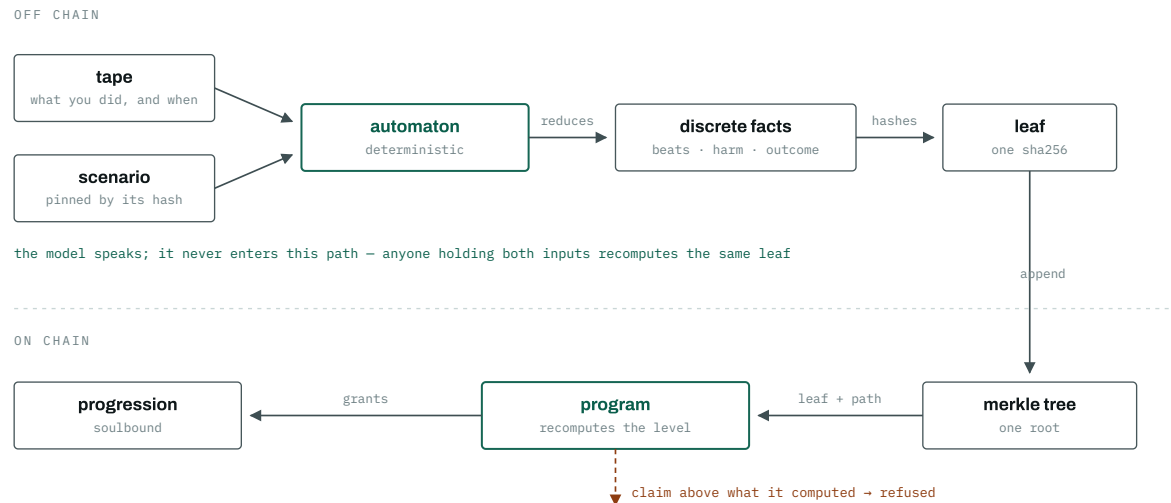
Decides transitions and endings. Exact and replayable.

THE PROOF

No model in proof path. Settles by arithmetic via \$VIGIL.

ZERO-TRUST ARCHITECTURE

Two inputs in, one hash out. The program refuses its own user.



The tape and the scenario are the only inputs. The chain never sees a run — only a leaf that has to prove against a root it built itself.

LIVE VALIDATOR · CLAIM: PROFICIENT

REJECTED — claimed Proficient, computed Competent (3 distinct attempts)

LIVE VALIDATOR · CLAIM: COMPETENT

GRANTED — stored Competent, xp 35. Stored level is what program computed.

No issuer key. No oracle. The Solana program recomputes the level from Merkle proofs and writes only what its own arithmetic agrees with.

ONCHAIN ARCHITECTURE

Four primitives, none of them invented here.

LAYER	WHAT IT DOES	SOLANA PRIMITIVE
Scenario registry	Authors publish; content stays off-chain, only hash + price + royalty split onchain	Anchor program · Token-2022
Replay anchor	One compressed leaf per run — the high-volume write	Bubblegum state compression
Progression	Levels, skill trees, badges — minted permissionlessly, predicate recomputed onchain	Token-2022 <i>NonTransferable</i>
Credential	Accredited issuer attests an EPA competency to the player's wallet	Attestation Service

VOLUME

A single cohort year across the World Directory is hundreds of millions of runs. Compression puts that at **~\$110 per million**. Anchoring only the exams would be the cheap-chain compromise and would kill the thesis.

LATENCY

The skill tree advances in the same screen as the outcome. On a slow chain that becomes a pending spinner and stops feeling like a result.

ONBOARDING

Fee-payer relay. A player never sees a seed phrase, never holds SOL, never learns any of this is happening.

The token secures verification. It never represents competence.

Someone has to replay the tape and attest the outcome. Make that one party and you have rebuilt the database we are trying to escape. **\$VIGIL is what makes verification permissionless and expensive to lie in.** A vigil is kept beside the patient, never of the patient — which is exactly where this token sits relative to the credential.

THE LOOP

STAKE

Operators bond \$VIGIL to accept replay work. Bond size sets how much throughput they can take.

VERIFY

Replay is deterministic, so **two honest verifiers must agree**. Fees are paid by institutions and sponsors — in USDC, never by players.

DISPUTE → SLASH

Disagreement is provable: anyone posts a bond and re-runs the tape. The wrong verifier is slashed, the challenger is paid from it. Fraud proofs work here because the computation is exact.

SINKS · SOURCES

sink	verifier bond (locked while active)	supply out
sink	scenario listing + dispute bonds	supply out
sink	slashing — burn share	burned
source	verification fees → operators	earned
source	replay royalties → scenario authors	earned
note	authoring is cheap, so the author market is real	—

NEVER TOKENIZED

- **Credentials, badges, skill trees.** Token-2022 NonTransferable, no exceptions — a tradeable "Expert in Cardiology" is credential fraud with extra steps.
- **Access to play.** No token gates a student out of practice, ever. Players never hold \$VIGIL and never see it — the arena has its own name, **3R**, so the two layers do not even share a word.

Allocation is a proposal, not a commitment, and no token ships inside the hackathon window: the verification market only has to exist once verification is worth attacking. Distribution weighted to operators and scenario authors — the two parties whose work the network actually consumes.

THE MARKET AROUND THE RECORD

Everyone can check it. Nobody can buy it.

Two questions the rest of the deck leaves hanging: who would actually run a verifier, and what happens when somebody just wants to pay their way in.

WHO CHECKS IT — AND WHAT CHECKING EVEN MEANS

THE ONE WHO RELIES ON IT

A residency programme about to trust the record **does not want us to be the only ones checking it.**

THE SCHOOL THAT SENT THEM

Wants its graduates' records to hold when someone else looks.

THE SPONSOR

Pays against a level, so would rather check it than take it on faith.

AN OPERATOR, FOR THE FEE

Arrives last. A laptop is the machine.

A verifier loads the same scenario, replays the same tape, compares beats, harm and outcome. **It judges nothing** — which is why two honest verifiers cannot disagree. **On day one there is one node and it is ours**; an open set matters once verification is worth attacking.

WHAT MONEY CAN OPEN — AND WHAT IT CANNOT

SKILL · ALWAYS FREE

Clear one, the next opens. No money, no wallet, no sponsor needed for the whole season.

A SPONSOR

Opens an episode for a cohort. The author is paid per replay; the player transacts nothing.

IMPULSE · CONSUMER ONLY

Skip the grind. Harmless, because of what it does not buy.

 **None of them buys standing.** Your level is recomputed from anchored runs, and the program neither knows nor cares how a scenario came to be unlocked. **You cannot buy a leaf — only the chance to make one.**

MARKET AND PRECEDENT

This shape already won this hackathon once.

FRONTIER 2026 · GRAND CHAMPION

CrowdBrain: *"Train in simulation → Qualify through QA → Route to real work."*

The same proven shape, built for clinicians instead of robots. Out of 2,857 submissions from 150 countries.

SCENARIO CREATION ENGINE · SKALD

Automated Medical Fact-Checking: AI showrunner scaffolds scenarios; clinical crew gates content.

Skald verifies the content. Solana verifies the play.

WHY MEDICINE FIRST

Highest stakes & regulation. Solve trust here and it generalises downward.

WHO PAYS

Institutions on standard invoices, sponsor scholarships. Players pay \$0.

HEAD START (2 MOS LIVE)

**30 Medical Institutions (>75% of TH) ·
290+ Clinicians (68% active) · 1st Place
@ NECTEC 2026.**

TEAM & ROLES

**Medical Sim Lead (Embla) × Solana Core
Rust Builder.** Live at
embla.megawiz.co.th.

WHERE IT STANDS, AND WHAT'S NEXT

Two halves built. One week to join them.

WORKING NOW (LIVE IN PRODUCTION)

- **Replay** → **leaf**. EP1 runs against shipped automaton; three tapes, three distinct leaves, reproducible across runs.
- **Program** → **refusal**. Native Solana program live on validator, recomputing levels and rejecting invalid claims.
- **Award-Winning Engine**. 1st Place @ NECTEC AI for Thai 2026 (team01.aiforthai.in.th/embla.megawiz.co.th).
- **Rust end to end**. One scoring implementation compiled into game, verifier, and onchain program.

THE REMAINING FOUR WEEKS

- **W1** — join the halves: `claim_progress` takes merkle proofs against anchored leaves instead of attempts handed to it.
- **W2** — compressed anchors, scenario registry, public verify page.
- **W3** — soulbound progression, badge-gated scenarios, sponsor escrow paying out on a provable level.
- **W4** — mainnet-beta, demo video, open-source release.

Cut on purpose

ZK selective disclosure · verifier DePIN with live staking · token launch · mobile · new clinical content. Each is a real v2 line and none survives four weeks. The credential layer stays a stub until an accrediting institution is standing next to it — progression doesn't need one, which is why progression is the demo.

The 60-second version: you play a patient who is dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.